

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed **single line diagrams (SLDs)**, wiring diagrams, and panel layouts for low-voltage switchgear, ensuring adherence to **IEC 61439** standards.
- Performed critical electrical calculations, including **load calculations**, short-circuit analysis, and **busbar sizing**, to validate design integrity and safety margins.
- Selected and specified key components such as **circuit breakers**, protection relays, and contactors, integrating them into comprehensive panel designs.
- Collaborated closely with **mechanical design teams** to ensure seamless integration of electrical schematics with panel enclosure layouts, reducing integration rework by **15%**.
- Prepared detailed **technical documentation** including Bill of Materials (BOM), test procedures, and operation & maintenance manuals, supporting smooth project handovers.
- Supported factory acceptance tests (FAT) and site commissioning activities, troubleshooting design issues and optimizing performance for cost and safety parameters.

SKILLS

Design Tools – AutoCAD Electrical, EPLAN

Electrical Design – Switchgear Design Principles, LV Systems, Power Distribution Systems, Protection and Control Systems

Standards & Compliance – IEC 61439, IEC 60439, NSAI IS10101:2025

Electrical Calculations – Load Calculations, Short-Circuit Analysis, Thermal Analysis, Busbar Sizing

Documentation – Single Line Diagrams (SLDs), Wiring Diagrams, Schematics, Bill of Materials (BOM), Test Procedures

Component Specification – Circuit Breakers, Busbars, Protection Relays, Contactors

Collaboration – Mechanical Design Teams, Manufacturing Teams, Project Managers

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

LV Switchgear Panel Design Automation

- Developed automated scripts using **Python** to generate initial drafts of **SLDs and wiring diagrams** based on component lists, reducing design setup time by **30%**.
- Integrated component selection logic with a central database, ensuring accurate and up-to-date specifications for **circuit breakers and protection relays**.

- Designed a modular framework for creating **Bill of Materials (BOMs)** and basic test procedures, improving consistency across multiple projects.

Power Distribution System Analysis Dashboard

- Built an interactive dashboard using **Power BI** to visualize **load calculations and short-circuit analysis** results for industrial power distribution systems.
- Developed complex **DAX measures** for performance analysis and trend identification, enabling stakeholders to quickly assess system capacity and fault current levels.
- Connected to various data sources via **Power Query**, transforming raw electrical data into actionable insights for system optimization and planning.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)